Seasoned doctors make clinical judgements based on their individual experiences, in addition to the relevant guidelines, and these judgements are generally accepted. A hypothetical AI system with the same “error rate” as physicians on such judgements would be far more scrutinised. One reason we naturally distrust machines is their lack of auditability, and in turn, accountability. “A comfort in human errors that can be followed and safeguarded against.” Physicians can legally be held to a reasonably competent standard based on notes explaining their decisions. Explainable AI has poorly mimicked it with unfaithful post-hoc chain-of-thought or by relying on attention weights. Additionally, ML, at its core, requires often unspoken assumptions that the proxies we use everywhere in training are good enough fits to the real-world values.

I’m currently working on the unspoken assumptions aspect by applying causal inference to provide actionable advice regarding patient discharge decision. Forcing the assumptions out during causal discovery and debating the DAGs with experts. World models are another possible direction to tackle it, with NYU being one of the best places to work on them. However NYUTron actually caught my eyes first. Actual access to entire EHRs is a huge step forward for research. It’s extremely rare to have collaboration at this scale and ease with a medical school, even under the same university. However associations and scores don’t fit well into interventional clinic workflows. The first benefit of causal inference is the ability to visually explain what the model believes. Applications could be via generated counterfactuals that can’t be created with the same confidence through pure ML. UX friendly communications that if we don’t monitor them until such conditions are met, they have a percentage higher chance of being readmitted or long-term patients. The invisibile thinking and considerations physicians wrestle with turned into a concise discharge note within an EHR.

(Startup story and final goals)Taking entrepreneurship and later health innovation master's courses will help me plan research that really makes it into practice. I tried launching my own startup targeting the national medical non-emergency helpline. Hearing Swedish folk complain about the lack of continuity within their health journeys, I pitched a medical AI transcript-to-note generation system to tackle the first and often poorly documented point of contact. Besides training me to find solvable real-world problems, it summarises my problem-solving approach, bringing us to my goals. I'm indifferent to the methodologies I pursue and the application domain. My criteria for research worth pursuing are that we investigate the root cause first and that the goal directly benefits the stakeholders involved. Not just hypotheticals and expecting someone else to find a purpose for it.

Going back further, my first professional experience as a data analyst at a consultancy is still the foundation of my analytical approach. The human aspect of understanding each stakeholder's perspective informs good technical decisions and often requires digging. It came into play when I was working on a simple dashboard with a hospital later. Even specialists in the same field disagreed on the data annotation and later visualisation, an experience that prepared me to work on DAGs in my current role. The reason I mention it now is that one finding from NYUTron was that local fine-tuning is the better approach in the end. In the dashboard project, there were certain complex surgeries that only a single surgeon performed for years. While an extreme example, I've been thinking of how to handle a physician's personal biases(good and bad) in AI training. Perhaps a controversial topic, but if we were to include the individuals working that shift, financial incentives, bed capacity, etc., we could investigate and accommodate them. Healthcare is a highly political topic, and our research should investigate all leads equally if the goal is to produce impartial results.